Smart Message Prioritization Using Multi-Level Text Urgency Detection

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Abstract-

Effective message prioritization is critical to achieving efficiency in current communication systems where the message recipients are flooded with a large number of short text messages. Although most current solutions view the problem of urgency detection as a two-class classification problem, current communication practices involve a more subtle definition of urgency. This article proposes a multi-level model of urgency detection where the incoming messages are labeled into the "Low," "Medium," and "High" level of priority as defined explicitly using time-oriented criteria. A labeled dataset of short and informal text messages was developed manually. Traditional machine learning algorithms such as Logistic Regression, Decision Tree, and Random Forest were trained using TF-IDF features extended by features more suitable to the tasks of urgency detection. The experiment shows the efficacy of the aforementioned models, where the Logistic Regression model emerged as the one achieving the highest macro F1-score value. The importance of time-oriented features as well as features of message styles to the task of urgency detection is highlighted by the analysis of the extent to which each feature set impacts the model's prediction result. A RESTful API implementation is provided as a proof of the feasibility of the proposed method.

Keywords- Smart message prioritization, urgency detection, text classification, multi-class classification, machine learning, explainable AI

I. INTRODUCTION

Along with the rising number of online communications that take place via email, messaging services, and collaborative tools, the need for efficient message prioritization is becoming one of the most important challenges. This is because the user is often confronted with a number of short messages, most of which are not necessarily urgent while others require immediate attention. This often results in missed deadlines and poor productivity levels if these messages are not recognized as soon as they are urgent. This is where the need to recognize the urgency of these messages arises.

Current research efforts on message prioritization and urgency detection, as it has been formulated so far, has been almost exclusively a binary classification problem. Messages have been classified as either urgent or not urgent. This is because urgency has been assumed to be a binary choice, which fails to properly represent reality. Messages that need to be processed by end of day require a level of urgency that messages that need to be processed immediately require. However, these messages would fall under the 'urgent' category, which, if this binary classification system was adopted, would fall under a similar category.

Another recent trend is that these models employ complex deep architectures like recurrent networks and transformer networks. Even though these models can yield a very good predictive accuracy, sometimes these models lack explainability. In many applications like message prioritization, models that require little computation and can explain their outcome can still perform an important task.

In response to these findings, this paper aims to propose a multi-level urgency detection framework for smart message priorities. The objective is cast as a three-class classification problem that classifies messages into Low, Medium, and High levels of urgency based upon explicit criteria that are specifically time-related. In this work, we will specifically focus on small, informal text messages that are typical for messages we find in everyday communication, such as work-related messages. Additionally, a manually annotated data set based on strictly defined levels of urgency will be developed.

The paper uses classical machine learning models, including Logistic Regression, Decision Tree, and Random Forest, together with TF-IDF-based lexical features and urgency-oriented linguistic features. This design choice permits a balancing of performance, interpretability, and computational efficiency. Besides, an explainability analysis is conducted to understand the influence of temporal expressions and stylistic cues on model predictions. A prototype deployment using RESTful API further demonstrates the practicality of the proposed approach.

The main contributions of this work are three-fold:

1. Framing Urgency Detection as a multi-class classification problem to reflect real-world communication requirements;
2. Creation of a time-rule-based annotated dataset of short messages; and
3. Elaboration of an interpretable and deployable machine learning framework for smart message prioritization.

II. LITERATURE REVIEW

A. Overview of Previous Studies

The early studies related to message priority were mainly related to email categorization, with the objective of finding important or urgent messages from a huge number of routine communications. The common machine learning methods like Naïve Bayes, Support Vector Machines, and decision trees were employed, with features like lexical features, composition, and email metadata. The results demonstrated that words related to urgency, as well as syntax features, can appropriately determine the priority related to structured communications.

This line of research was later extended to other text classification tasks, including notification filtering and task priority, to name a few. Yet, most of the literature followed suit in reducing the urgency detection problem to a simple binary text classification problem, where the message is classified into the categories of urgent and non-urgent messages respectively. This simplification, although helpful in training and testing the model, did not go well with the inherent variability and nuances involved in the problem of urgency detection.

Later approaches have focused on deep learning models such as recurrent neural networks and transformer networks that aim to address semantic and contextual aspects of the text. These models have outperformed older models on large corpora of text but are often computationally expensive and require large annotations of data. At the same time, they are less interpretable and make it difficult to develop applications such as message prioritization that require an understanding of the underlying logic behind the predictions.

B. Underlying Theory

The task of urgency detection can also be considered a text classification problem, wherein linguistic and structural patterns in the text are used to determine the time sensitivity of the text. On a theoretical level, the issue of urgency in text can be identified by the use of temporal language, imperative sentences, and stylistic features such as capitalization or punctuation.

Conventional vector space representations, such as TF-IDF, capture text using the concepts of importance and co-occurrence, allowing classifiers to discover characteristic lexical cues that are indicative of urgency. Jointly used with interpretable classifiers like Logistic Regression and Decision Trees, such models enable both accurate prediction as well as interpretable reasoning. This reflects the theoretical underpinning for using lightweight and feature-driven models for detecting urgency, especially in cases with small and informal text messages.

C. Research Gaps

Despite active research in message prioritization, several gaps still remain: most of the existing studies formulate urgency detection as a binary classification problem, which inadequately represents real-world communication demands when urgency can come in multiple degrees; there is increasing dependence on complex deep learning models that are often powerful yet sacrifice interpretability or even feasibility in deployment; and most datasets do not have explicit and reproducible criteria for annotating urgency, which results in subjective labeling and inconsistent evaluation.

Furthermore, relatively little attention has been paid to short, informal messages commonly exchanged through instant messaging platforms, where urgency cues are more subtle and stylistic. The current work addresses these lacunae by proposing a multilevel urgency detection framework resting on clear, time-oriented definitions and interpretable machine learning models. In line with this perspective, this paper contributes to a workable and transparent solution for smart message prioritization while keeping the predictive performance strong.

III. PROBLEM FORMULATION AND DATASET

This section describes the formulation of the urgency detection problem, the data collection process, and the dataset characteristics used in this study.

A. Problem Formulation

Smart message prioritization is addressed as a multiclass text classification problem. Given a small text message m, the system should label it according to one of three degrees of urgency: Low, Medium, and High. Three levels of urgency are introduced grounded on explicit time-related criteria to meet realistic communication needs.

- Low Priority: Messages with casual or informational content and placing no explicit time constraint.
- Medium Urgency: The messages that require action at a future time, usually starting from the next day onwards, but do not demand the same-day attention of the recipient.
- High: These are messages necessitating action on the same day or requiring immediate attention, whereby delays would incur adverse results.

Let the feature vector be represented for every message as X and the urgency label as $y \in \{0,1,2\}$, where increasing numeric value corresponds to an increase in the urgency level.

B. Data Collection

The dataset compiled for this research was created manually to reflect brief, less formal messages that one would exchange in day-to-day communication situations: things like reminding a fellow of an upcoming meeting, updating one on the status of a certain task, or simply letting someone know what's going on. The messages were developed to be similar to text messages or instant messaging and ranged in length from 5 to 30 words.

Manual collection of data was the best option in allowing for precise control over the message structure and urgency cues; meanwhile, consistency with explicitly defined urgency categories is maintained. Another important reason for the above-mentioned choice is that this avoids ethical and privacy concerns due to the use of real user-generated communication data. Messages were created to cover a variety of contexts and linguistic styles, which allowed capturing diverse urgency-related patterns.

C. Urgency Annotation Rules

We assigned urgency labels based on overt time-based annotation rules to reduce subjectivity and enhance the reproducibility of the approach. Temporal constraints were annotated irrespective of emotional tone and message politeness.

- Messages containing no explicit mention of a deadline or time were labeled as Low urgency.
- Messages specifying a deadline beyond the current day -e.g., "tomorrow" or later-were labeled as Medium urgency.
- Messages needing same day action or with immediate temporal cues "today", "now", "ASAP" were classified as High Urgency.

If it was ambiguous between two urgency levels, the message was conservatively assigned to the lower urgency class. This rule was introduced in order to not overestimate urgency and guarantee consistency of labeling.

D. Dataset Characteristics

The final dataset involves a few hundred labeled messages spread out through the three levels of urgency. The class distribution was intended to be fairly balanced. This reflects a realistic situation, as in actual communication processes. The data involves a text message and the label concerning the level of urgency.

The dataset pertains to small, informal text messages and not to longer or more structured forms of documentation, which makes the dataset apt for testing the urgency detection capability within the bounds of real-world messaging applications.

IV. METHODOLOGY

A. *Overview of Proposed Framework*

A supervised learning framework is followed in the scheme, which is implemented using a manually labeled dataset for short text messages, based on which specific urgency levels are defined. The messages would be processed using machine learning, which would be implemented after structuring the messages, followed by the training phase where models would be developed for classifying the messages as belonging to any one of the three categories, namely Low, Medium, or High.

B. Text Preprocessing

Text preprocessing is carried out to remove variations of messages so that there is less variability and less noise. It also maintains urgency-related information. Text preprocessing involves making all letters of the message lowercase and removing all non-alphabetic characters. Stopwords are filtered out as they are frequent words that don't add any significant information regarding the urgent classification of messages.

Stemming and lemmatization are intentionally avoided in order to preserve original word forms, especially temporal phrases that play a vital role in urgency analysis. The same preprocessing process is followed whenever the training, testing, or deployment phase is executed.

C. Feature Engineering

To capture both semantic content and stylistic urgency cues, a set of lexical-linguistic characteristics is used.

1) *Lexical Features:* Lexical features are extracted using the Term Frequency-Inverse Document Frequency (TF-IDF) method, considering unigrams and bigrams. The TF-IDF vector representation encompasses discriminative words and short phrases related to the urgency, specifically the explicit temporal phrases used in the process of annotating the dataset.
2) *Urgency-Oriented Linguistic Features:* Apart from TF-IDF variables, the model includes four linguistic variables specific to urgency:
   - Message length measured as the number of words
   - Number of exclamation marks
   - Binary feature for the presence of keywords related to urgency

- Proportion of capital letters to total characters

Such properties include the characteristic use of structural and stylistic elements of urgency in informal communication.

To form the final feature vector for each message, the TF-IDF feature matrix is concatenated with the linguistic feature set.

D. Classification Models

Three traditional machine learning algorithms are tested in relation to their applicability in multi-level urgency detection.

1) *Logistic Regression:* Logistic Regression acts as a base level multi-class classifier because of its excellent generalization performance. Multinomial logistic regression allows direct calculation of class probabilities, making predictive as well as explainability analyses easier.

2) *Decision Tree*: To extract relationships between feature sets and levels of urgency in a rule-based manner, the Decision Tree classifier is used. Using recursive division of the feature space, the classifier derives interpretable rules that correspond well with time-defined levels of urgency.

3) *Random Forest*: Random Forest, which is an assortment of decision trees, is employed to represent the non-linear relationships between lexical/linguistic variables. This is because Random Forest reduces overfitting as well. Moreover, it is capable of determining variable importance.

E. Model Training and Evaluation Protocol

All models are trained using the same feature representation and ordinal label encoding corresponding to the urgency levels defined in Sec. III. The dataset is split into a training and a test set using stratified sampling to preserve class distribution.

Precision, recall, and F1-score for each urgency class are used to assess the performance of the models. Macro-averaged metrics are used to ensure that there is a balanced evaluation in respect of the urgency level, assuming there may be class imbalance. Confusion matrices are investigated to find out patterns in misclassifications, especially between neighboring classes in terms of urgency.

F. Explainability Strategy

The key motivation of the model proposed in this paper is explainability. These findings are then validated through Logistic Regression by analyzing model coefficients to identify those features that positively or negatively influenced each urgency level and highlighting the dominant urgency indicators through Random Forest feature importance scores.

It also generates example-based explanations by looking into the prediction confidence scores for the selected test messages. Qualitative analysis provides insight into model behavior and supports transparency in urgency prediction.

G. Prototype Deployment

In order to check the applicability of this model, a prototype RESTful API has been developed using FastAPI in a cloud-based notebook environment. This prototype takes a short text message as input and provides a predicted level of urgency along with class-specific confidence scores as output. This prototype demonstrates how this methodological approach could be used within a real-world setting for text message prioritization.

V. EXPERIMENTAL SETUP

A. Data Splitting Strategy

The dataset was divided into training and test sets by stratification so that the urgency class distribution is preserved. A fixed train–test split was employed, where the majority of samples are used for training the model while the rest are reserved for evaluation. Stratification ensures that all urgency levels are well represented in both subsets to allow for a proper, trustworthy performance assessment.

B. Implementation Details

All experiments were carried out using the Python programming language and its standard machine learning libraries. TF-IDF vectorization was applied to the preprocessed text to generate lexical features, which were subsequently concatenated with urgency-oriented linguistic features. Model training and evaluation were done in a cloud-based notebook environment to ensure reproducibility.

To keep consistency in experiments, identical preprocessing steps, feature representations, and label encodings were applied to all models. The selected hyperparameters are conservative, balancing model performance and generalization without extensive tuning.

C. Evaluation Metrics

Performances of models were evaluated in terms of precision, recall, and F1-score for each urgency class. Along with these, the results include macro-averaged precision, recall, and F1-score to account for a potential class imbalance, so that all the urgency levels are weighted equally in describing the performance.

Confusion matrices were generated in order to analyze the pattern of misclassifications, especially among adjacent urgency levels. Accuracy was reported as a supplementary metric, but not used as the main indicator of model performance due to the multi-class nature of the task.

VI. RESULTS AND DISCUSSION

A. Quantitative Results

Table I provides a summary of the performance of the models evaluated, in terms of macro-averaged precision, recall, and F1-score. Among the three models, Logistic Regression has the highest macro-averaged F1-score, showing that it balanced its performance quite well on all urgency classes. Random Forest follows closely behind, while the Decision Tree performed comparatively poorly, especially on recall.

Table I. Performance Comparison of Urgency Detection Models (Macro-Averaged Metrics)

Model	Precision	Recall	F1-score
Logistic Regression	1.00	0.98	0.99
Decision Tree	0.96	0.96	0.96
Random Forest	0.99	0.97	0.98

These results indicate that linear decision boundaries are adequate to capture the urgency-related patterns when combined with well-defined lexical and linguistic features in short text messages. Strong performance exhibited by Logistic Regression further underlines the effectiveness of interpretable, low-complexity models for this task.

B. Class-Wise Performance Analysis

Class-wise evaluation reveals that High urgency messages are classified with consistently high precision and recall across all models. The explicit temporal expressions such as same-day deadlines present in High class may provide the major discriminating context that differentiates this class from other classes in a very strong way.

Medium urgency messages are relatively harder to classify, as they tend to share lexical similarities with Low and High urgency messages. Nonetheless, Logistic Regression and Random Forest remain robust, suggesting that the combination of TF–IDF features and urgency-oriented linguistic features is effective at capturing future-oriented temporal cues.

Low urgency messages are also classified reliably, with little confusion noticed with High urgency messages. This attests to the fact that the defined annotation rules and feature design effectively distinguish non-time-sensitive communication from urgent content.

C. Confusion Matrix Analysis

Fig. 1 presents the confusion matrix for the Logistic Regression model on the test dataset. The results show high accuracy in classifying instances in all levels of urgency. The high-level urgency messages are correctly classified with perfect accuracy because they all have been correctly identified as High urgency messages. There are no errors in these classifications.

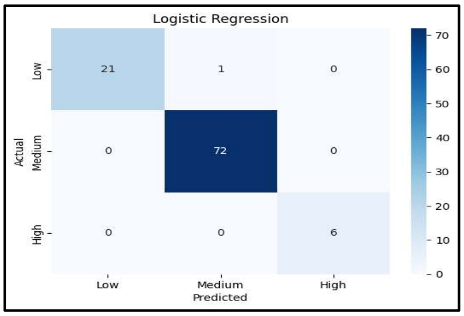

Fig. 1. Confusion matrix for Logistic Regression showing classification performance across Low, Medium, and High urgency levels.

In the case of Medium urgency messages, the model obtains perfect separation, where all the messages are correctly predicted, without being confused either with the Low or High urgency messages. This implies that the model can clearly distinguish based on the annotation rules that are future-oriented.

There is a little confusion found in Low urgency messages. Here, one example is labeled as Medium urgency. It is probably due to messages that have a preponderance of information with mild cues of time or tasks that overlap with the Medium urgency patterns. It is important to note that there is no Low urgency message that is labeled as High urgency.

In any case, it can be seen that the overall confusion matrix confirms a realistic restriction of misclassifications to adjacent urgency values, while the fact that there are no confusions for the Medium versus High urgency classes confirms the efficacy of the proposed time urgency measures.

D. Discussion

In conclusion, the results show that even without adopting more complex deep learning models, the proposed framework is very effective. The success of Logistic Regression indicates that urgency detection, especially for short, informal communications, relies primarily on overt urgency and style features that can be represented by light-weight features.

Even if high performance metrics suggest a successful task of classification, an interpretive evaluation implies certain weaknesses in dealing with implicitly time-sensitive messages. Such a result tends to support a trade-off not only in interpretability but in semantic meaning as well.

VII. EXPLAINABILITY ANALYSIS

Explainability is one of the main objectives of the proposed urgency detection model, especially when considering its usage for message prioritization applications. For understanding the impact of various variables on the predictions made by the model, the analysis for feature importance was performed.

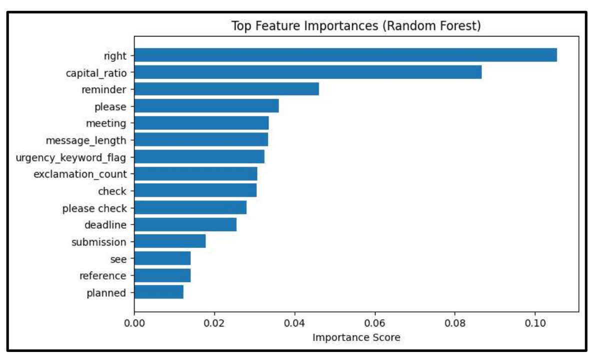

Fig. 2. Top feature importances derived from the Random Forest model, highlighting lexical and stylistic indicators relevant to urgency detection.

Figure 2 shows the most important features imported by the Random Forest model. These results demonstrate the contribution of both lexical and stylistic features to the urgency detection task. Amongst the many features assessed, “right,” “reminder,” “deadline,” and “submission,” and other action-oriented words related to time appear to play an important role with high values for the importance score, as would have been expected based on the clear time-based criteria.

Stylistic features are also given considerable weight. The ratio of capital letters comes out to be one of the most prominent features, which reveals that the style of writing with capital letters conveys a crucial signal to make a decision about urgency. Just like this, the length of the message, number of exclamation marks, and the presence of the keyword ‘urgency’ also impact the decision-making process.

Interestingly, the polite/task-related vocabulary like ‘please,’ ‘meeting,’ and ‘check‘ is seen among the crucial elements. These words, by themselves, do not convey a sense of urgency, yet they tend to be associated with messages that have actions linked to them and possibly have some notion of the timeline associated with that action.

In sum, the feature importance analysis shows the importance of feature learning in the context of the problem at hand and indicates the meaningfulness and interpretability of the models' learning for the problem domain based on real-world communication behavior. The high importance level of temporal expressions and stylistic features underscores the correctness of feature design and the usefulness of lightweight models for the problem.

VIII. SYSTEM DEPLOYMENT

To show the real-world applicability of the proposed urgency detection framework, the trained model was deployed as a prototype using the FastAPI framework for building RESTful APIs. The deployment serves to act like a proof-of-concept to show how the system can be integrated into real-world message prioritization pipelines.

This API takes as input a small message and returns the predicted urgency level, which could be one of Low, Medium, or High, along with class-wise confidence scores. Similarly, the deployment pipeline mimics the training setup in that exactly the same preprocessing steps are carried out, together with feature extraction procedures during model development, mirroring the training-inference consistency.

The prototype thus implemented and tested within the cloud-based notebook environment allowed agile experimentation and reproducibility without the need for any special server infrastructure. This lightweight deployment approach demonstrates how interpretable machine learning can be used in a messaging system to assess urgency in real time.

While this deployment currently is used for demonstration purposes, the API design is modular, allowing easy extension for production environments or integration with messaging platforms, notification systems, or task management tools.

IX. LIMITATIONS AND FUTURE SCOPE

Although the proposed multi-level framework for urgency detection shows strong performance and interpretability, there still exist some drawbacks. Firstly, the approach considers the individual level of messages and does not use the conversation or historical background. Therefore, the escalation of the urgency level based on the sequences of messages as well as the subsequent reminders is not considered. Hence, the level of the messages with implicitly time-sensitive intent but with no explicit time expressions might be undervalued.

Secondly, the dataset for the research is annotated by hand, which allows exact regulation of the annotative criteria, yet does not possibly reflect the complete linguistic variability as seen in natural communication. Despite the fact that the dataset represents typical patterns for informal messages, perhaps an even larger dataset could improve robustness further.

Future studies can further improve the model by incorporating context-aware features, such as the previous messages, time duration between messages, and semantic representations involving pre-trained language models, into the model, thereby capturing the progression of urgency more efficiently. Extending the dataset by incorporating real anonymized messages and the model's performance evaluation within various domains of communication can also be promising areas for future studies.

X. CONCLUSION

A framework for urgency detection was provided for efficient message prioritization by machine learning algorithms that support interpretations. In this research, urgency detection was framed as a three-class problem, which is more feasible, as opposed to binary urgency detection, which has limitations due to rules defined for annotation.

Experiment outcomes show the effectiveness of classical machine learning approaches, especially Logistic Regression, using TF-IDF features and linguistic signals of urgency, and explainability report verifies the importance of temporal expressions and style indicators in the process of urgency modeling, ensuring the validity and practicability of the solution design.

The whole solution is further applied in a RESTful API-based prototyping environment, verifying the practicability of integrating this solution system within actual short messaging applications or services.

In summary, this study proves that lightweight machine models with interpretability capabilities are useful in the smart message prioritization process given effective alignment of problem formulation and feature design with real-life communication requirements.

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